

JUSTIN PAYAN

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[LinkedIn](#) ♦ [Github](#) ♦ [Google Scholar](#)

ACADEMIC POSITIONS

Carnegie Mellon University

Oct '24 - Present

Postdoctoral Research Associate with [Nihar Shah](#)

Pittsburgh, PA

- Applying large language models to improving scientific reasoning and peer review
- Developing dynamic scheduling algorithms for ground-based telescopes

RESEARCH INTERESTS

I optimize scientific productivity using techniques from operations research, data science, machine learning, and natural language processing.

EDUCATION

University of Massachusetts Amherst

Sept '18 - February '25

PhD Computer Science

(advised by [Yair Zick](#))

University of Georgia

Sept '13 - May '18

M.S. Artificial Intelligence, B.S. Mathematics, A.B. Cognitive Science

(advised by [Frederick Maier](#))

INDUSTRY EXPERIENCE

Adobe Research, Document Intelligence Lab

May - August '23

Research Intern under [Chris Tensmeyer](#)

College Park, MD

- Document structure inference using deep learning, computer vision, and combinatorial optimization

Microsoft, Calc Intelligence

June - September '22

Data Science Intern under [Kartik Sridhar](#)

Redmond, WA

- Data augmentation using GPT-3 for an NL to code task
- Worked closely with both production and research teams to implement state-of-the-art modeling and evaluation

Amazon Alexa, Trustworthy Alexa

June - September '21

Research Intern under [Yuval Merhav](#)

Cambridge, MA (remote)

- Applied generative insertion transformers (GIT) to data augmentation for NER
- Investigated interaction of GIT augmentation with knowledge-based augmentation approaches

Amazon Alexa, Trustworthy Alexa

May - August '20

Research Intern under [Yuval Merhav](#)

Cambridge, MA (remote)

- Explored the efficacy of generative replay for continual learning in a privacy-aware setting

- Created and released dataset for evaluating continual learning for single-task NER

Vertica, Machine Learning

June '17 - June '18

Software Engineer under Vincent Xu

Cambridge, MA

- Implemented distributed machine learning algorithms in SQL and C++, including k-means++
- Designed, built, and maintained data preprocessing functions, such as one-hot encoding, normalization, and missing value imputation

PUBLICATIONS

$(\alpha - \beta)$ indicates alphabetical order; asterisks indicate equal contribution.

1. [greCAPTCHA: Assessing Understanding as Evidence of Research Authorship Under Generative AI](#)
Justin Payan*, Bálint Gyevnár*, Atoosa Kasirzadeh, and Nihar B. Shah
Under Submission
2. [ML Researchers Support Openness in Peer Review But Are Concerned About Resubmission Bias](#)
Vishisht Rao, **Justin Payan**, Andrew McCallum, and Nihar B. Shah
Under Submission
3. [FLAWS: A Benchmark for Error Identification and Localization in Scientific Papers](#)
Sarina Xi, Vishisht Rao, **Justin Payan**, and Nihar B. Shah
Under Submission
4. [Uncertainty-aware Ground-based Telescope Observation Scheduling at ALMA](#)
Justin Payan, Noemi Barbagli, Ignacio Toledo, Rodrigo A. Carrasco, Sergio Martín, and Nihar B. Shah
SPIE Astronomical Telescopes and Instrumentation 2026
5. [Fair and Welfare-Efficient Constrained Multi-matchings under Uncertainty](#)
Elita Lobo*, **Justin Payan***, Cyrus Cousins, and Yair Zick
NeurIPS 2024
6. [InstructExcel: A Benchmark for Natural Language Instruction in Excel](#)
Justin Payan, Swaroop Mishra, Mukul Singh, Carina Negreanu, Christian Poelitz, Chitta Baral, Subhro Roy, Rasika Chakravarthy, Benjamin Van Durme, and Elnaz Nouri
EMNLP 2023 (Findings)
7. [Into the Unknown: Assigning Reviewers to Papers with Uncertain Affinities](#)
Cyrus Cousins, **Justin Payan**, and Yair Zick ($\alpha - \beta$)
SAGT 2023
8. [Graphical House Allocation](#)
Hadi Hosseini, **Justin Payan**, Rik Sengupta, Rohit Vaish, and Vignesh Viswanathan ($\alpha - \beta$)
AAMAS 2023
9. [Relaxations of Envy-Freeness over Graphs](#)
Justin Payan, Rik Sengupta, and Vignesh Viswanathan ($\alpha - \beta$)
AAMAS 2023 (Extended Abstract)

10. [I Will Have Order! Optimizing Orders for Fair Reviewer Assignment](#)
Justin Payan and Yair Zick
IJCAI 2022
11. [Towards Realistic Single-Task Continuous Learning Research for NER](#)
Justin Payan, Yuval Merhav, He Xie, Satyapriya Krishna, Anil Ramakrishna, Mukund Sridhar,
 and Rahul Gupta
EMNLP 2021 (Findings)
12. [Online Post-Processing in Rankings for Fair Utility Maximization](#)
 Ananya Gupta*, Eric Johnson*, **Justin Payan**, Aditya Roy, Ari Kobren, Swetasudha Panda,
 Michael Wick, and Jean-Baptiste Tristan
WSDM 2021
13. [Document Representations using Fine-Grained Topics](#)
Justin Payan, Nicholas Monath, and Andrew McCallum
Sets & Partitions Workshop at NeurIPS 2019

CODE CONTRIBUTIONS

PULSAR 2026
 Designed the [PULSAR](#) telescope scheduling algorithm, which improves astronomical objective completion while improving satisfaction of global scheduling constraints. PULSAR is currently being evaluated for use at the ALMA Telescope.

OpenReview 2022
 Created and deployed the [FairSequence reviewer matching algorithm](#) on OpenReview platform, used by over 50 peer review venues.

AWARDS, GRANTS, & SCHOLARSHIPS

- Co-PI. Alfred P. Sloan Foundation. “Reliable Autonomous AI Scientists.”
 With Nihar B. Shah and Atoosa Kasirzadeh. 2026 – 2027.
- Google Cloud Computing Credits (Spring 2026)
 Amount: Equivalent of \$30,000
- UMass CICS Proposal Writing Grant (Spring 2023)
 Amount: \$12,733.80
- University of Georgia Foundation Fellowship (2013-2017)
 Amount: \$128,260 in tuition and stipend for housing, research, and travel

INVITED TALKS

- Brown University (E-GLAMOR Group), March 2023
 Harvard University (EconCS Seminar), January 2023
 Carnegie Mellon University (Nihar Shah’s Group), November 2022
(Into the Unknown: Assigning Reviewers to Papers with Uncertain Affinities)

- UMass Data Analytics and Computational Social Science Brownbag Series, April 2021
(*Envy-Freeness in Paper Reviewer Assignment*)
- UMass Amherst Theory Seminar, April 2021
(*Fair Reviewer Assignment*)

ACADEMIC SERVICE

- **Workflow Chair** for IJCAI 2023
- **Workshop Co-organizer** for [Computational Fair Division](#) at IJCAI 2023-2025, [Practical Improvements to Peer Review](#) at AAMAS 2025, [Workshop on Incentives in Academia](#) at EC 2025
- **Program Committee / Reviewer** for ACL 2023, EC 2025, KDD 2025
- **Subreviewer** for NeurIPS 2024, EC 2024, AAMAS 2022, GAIW at AAMAS 2022

TEACHING / MENTORING / OTHER SERVICE

- **Instructor of Record** at UMass Amherst for *Intro to Numerical Computing with Python* (Fall 2020 & 2021)
- **Graduate Teaching Assistant** at UMass Amherst for *Intro to AI* (Fall 2018), *Intro to Problem Solving with Computers* (Fall 2020), and *Advanced Algorithms* (Spring 2021)
- **Mentees:**
 - Rudra Gopal. BS, University of Massachusetts. Now MS at Georgia Tech.
 - Gautam Chatterjee. BS, University of Massachusetts. Now Software Engineer at Wayfair.
 - Ed Almusalamy. BS, University of Massachusetts. Now MS at University of Massachusetts.
 - Hassan Gudal. BS, University of Massachusetts. Graduated 2024.
 - Noemi Barbagli. BS, Carnegie Mellon University.
 - Eric Johnson. MS, University of Massachusetts. Graduated 2021.
 - Ananya Gupta. MS, University of Massachusetts. Now MLE at Diffbot.
 - Aditya Kumar Roy. MS, University of Massachusetts. Graduated 2021.
 - Vedant Bhatia. MS, University of Massachusetts. Now Quantitative Developer at HBK Capital Management.
 - Vishisht Rao. MS, Carnegie Mellon University. Now PhD student at Carnegie Mellon University.
 - Sarina Xi. MS, Carnegie Mellon University. Now PhD student at University of Wisconsin-Madison.
 - Arshika Lalan. MS, Carnegie Mellon University. Now Senior ML Scientist at JPMC.

SKILLS

Python, Gurobi, CVXPY, linear programming, integer programming, combinatorial optimization, machine learning, named entity recognition, large language models, Pandas, Numpy, Scikit-learn, Pytorch